CPSC 332 File Structures and Database Systems

Fall 2026 Term Project
©Dr. Shawn X. Wang [1]
Project Report Due: 12/18/2026

The Team Project:
The project is to be completed in groups. Each group has three students. The project will use MySQL database and PHP to build a web database application. Please visit the Canvas class website and click on "Check Your Accounts" to see the account and password assigned to you. The server for the project is located behind the firewall of the campus network. You need to come to campus using the Wi-Fi to access the server. If you would like to access the server from home, you must use VPN.

Here are what you need to turn in in the project report:
A PDF file of the project report that includes the following items. Upload a single PDF file to Canvas on the Friday of the final exam week.
1. A cover page listing the team member information (the names and last 4 digit of campus ID) and the account that has the project implementation. Since there are three members each with a different account, your team needs only to upload the project implementation into one account.
2. The ER diagram of your design (similar to homework #1).
3. The corresponding relational model with primary keys and foreign keys clearly specified (cf. Fig. 9.2 on page 291, similar to homework #2).
4. The source code of the project, including the DDL that you used to create the tables and the SQL that you used to implement the queries, as well as the PHP scripts that you used to implement the user interface.
5. The screenshots of the interfaces you developed.
6. The screenshots of sample runs of the system, including inputs and outputs.

The implementation of the project should be uploaded in one of the accounts that are given to you on the department server. Only projects that are completed and tested on the server will receive full credits. If the implementation can not be tested you will receive at most 80% of the credits.

Please find the Project Descriptions on the second page.

[1] *This document is to be used only by the students who are taking CPSC332 in Fall 2026. Sharing with anyone else is prohibited.*

Project Descriptions:
You are asked to design a database for a university that satisfies the following requirements:
1. The database keeps information of each professor, including the social security number, name, address, telephone number, sex, title, salary, and college degrees. The address includes street address, city, state, and zip code. The telephone number includes the area code and 7 digit number.
2. Each department has a unique number, name, telephone, office location, and a chairperson who is a professor.
3. Each course has a unique number, title, textbook, units. Each course also has a set of prerequisite courses. Each course is offered by a department.
4. Each course may have several sections. Each section has a section number that is unique within the course, a classroom, a number of seats, meeting days, a beginning time, an ending time. Each section is taught by a professor.
5. The database keeps student records, including the campus wide ID, name, address, and telephone number. Each student majors in one department and may minor in several departments. The name includes first name and last name.
6. The database keeps enrollment records. Each record has a student, a course section, and a grade.

Design your database and create at least the following numbers of records:

8 students	2 departments	3 professors
4 courses	6 sections	20 enrollment records

You should provide interfaces for the following individuals:

For the professors:
a. Given the social security number of a professor, list the titles, classrooms, meeting days and time of his/her classes.
b. Given a course number and a section number, count how many students get each distinct grade, i.e. ‘A’, ‘A-’, ‘B+’, ‘B’, ‘B-’, etc.

For the students:
a. Given a course number list the sections of the course, including the classrooms, the meeting days and time, and the number of students enrolled in each section.
b. Given the campus wide ID of a student, list all courses the student took and the grades.